FAST-scDECO

Flexible and adaptive scalable tool for single-cell differential co-expression

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Introduction

Bivariate observations:

$$(Y_{11}, Y_{12})$$
$$(Y_{21}, Y_{22})$$
$$(Y_{31}, Y_{32})$$
$$\vdots$$
$$(Y_{n1}, Y_{n2})$$

p-dimensional covariate tuples:

$$(x_{11}, x_{12}, \ldots, x_{1p})$$
$$(x_{21}, x_{22}, \ldots, x_{2p})$$
$$(x_{31}, x_{32}, \ldots, x_{3p})$$
$$\vdots$$
$$(x_{n1}, x_{n2}, \ldots, x_{np})$$

How is $\operatorname{corr}(Y_{i1}, Y_{i2})$ affected by $(x_{i1}, \ldots, x_{ip})$?

What is the distribution $f(y_{i1}, y_{i2} \mid x_{i1}, \ldots, x_{ip})$?

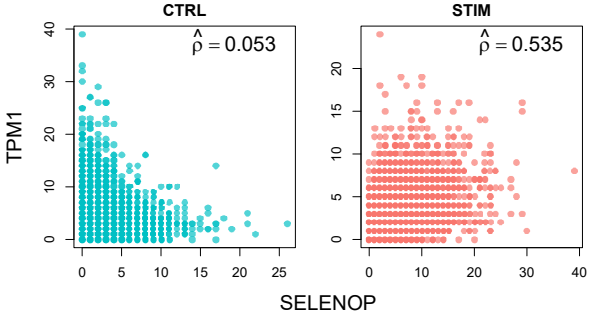
Introduction

Bivariate observations:

$$(\text{SELENOP}_1, \text{TPM1}_1)$$
$$(\text{SELENOP}_2, \text{TPM1}_2)$$
$$\dots$$
$$(\text{SELENOP}_n, \text{TPM1}_n)$$

factor covariate w/ 2 levels:

STIM
CTRL
...
STIM



Model Framework

A bivariate Gaussian copula is a good candidate for analyzing correlation in count data because it can model correlated data with arbitrary marginal distributions.

Bivariate Gaussian Copula PMF (Discrete Marginals)

$$c\left(y_1, y_2; \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \rho\right) = \int_{\Phi^{-1}\left\{F_1^{-1}(y_1-1;\boldsymbol{\theta}_1)\right\}}^{\Phi^{-1}\left\{F_1^{-1}(y_1;\boldsymbol{\theta}_1)\right\}} \int_{\Phi^{-1}\left\{F_2^{-1}(y_2-1;\boldsymbol{\theta}_2)\right\}}^{\Phi^{-1}\left\{F_2^{-1}(y_2;\boldsymbol{\theta}_2)\right\}} \phi_2\left\{t_1, t_2;\ \rho\right\} dt_2 dt_1$$

The ρ parameter can be modeled as a function of covariates,

$$\rho = \tanh\left\{\tau_0 + \tau_1 x_1 + \ldots + \tau_p x_p\right\},$$

so that hypotheses can be tested about how covariates such as cell type affect the gene-gene correlation.

Model Framework

For FAST-scDECO, we modify this Gaussian copula PMF to account for zero-inflation.

Let p_1, p_2 be the probabilities that an observation from marginal 1, marginal 2, respectively, gets a zero value from a drop-out event. Then any 0 we observe could have come from zero-inflation or it could have been produced naturally.

FAST-scDECO PMF (Discrete Marginals)

$$f(y_1, y_2 | p_1, p_2, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \rho) =$$

$$= (1-p_1)(1-p_2)c(y_1, y_2 | \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \rho) + \begin{cases} 0 & \text{if } y_1 \neq 0 \text{ and } y_2 \neq 0 \\ (1-p_1)p_2 f_1(y_1; \boldsymbol{\theta}_1) & \text{if } y_1 \neq 0 \text{ and } y_2 = 0 \\ p_1(1-p_1) f_2(y_2; \boldsymbol{\theta}_2) & \text{if } y_1 = 0 \text{ and } y_2 \neq 0 \\ p_1 p_2 & \text{if } y_1 = 0 \text{ and } y_2 = 0 \end{cases}$$

Model Framework

All parameters in the FAST-scDECO PMF can be made functions of covariates in a similar way to how we did it for ρ:

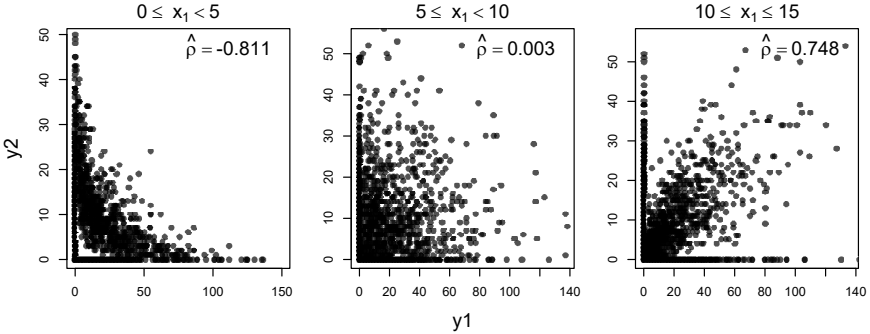
Parameters as functions of covariates

$$
\begin{aligned}
p_1 &= \text{sigmoid}\{\kappa_{01} + \kappa_{11}x_1 + \cdots + \kappa_{p1}x_p\} \\
p_2 &= \text{sigmoid}\{\kappa_{02} + \kappa_{12}x_1 + \cdots + \kappa_{p2}x_p\} \\
\mu_1 &= \exp\{\beta_{01} + \beta_{11}x_1 + \cdots + \beta_{p1}x_p\} \\
\mu_2 &= \exp\{\beta_{02} + \beta_{12}x_1 + \cdots + \beta_{p2}x_p\} \\
\sigma_1 &= \exp\{\alpha_{01} + \alpha_{11}x_1 + \cdots + \alpha_{p1}x_p\} \\
\sigma_2 &= \exp\{\alpha_{02} + \alpha_{12}x_1 + \cdots + \alpha_{p2}x_p\} \\
\rho &= \tanh\{\tau_0 + \tau_1x_1 + \cdots + \tau_px_p\}
\end{aligned}
$$

MLE is used to estimate the parameters of the model, $\boldsymbol{\kappa}_1, \boldsymbol{\kappa}_2, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \boldsymbol{\tau}$

Simulation

- We simulate $n = 5000$ realizations from unif$(0, 15)$ to act as covariate $\boldsymbol{x}_1$
- Specify $\kappa_1 = -0.93$, $\kappa_2 = -1.22$, $\beta_{01} = 3.16$, $\beta_{02} = 2.41$, $\alpha_{01} = -0.23$, $\alpha_{02} = 0.50$,
- Specify $\rho_i = \tanh\{-1.62 + 0.21x_1\}$
- Simulate (y_1, y_2) pairs

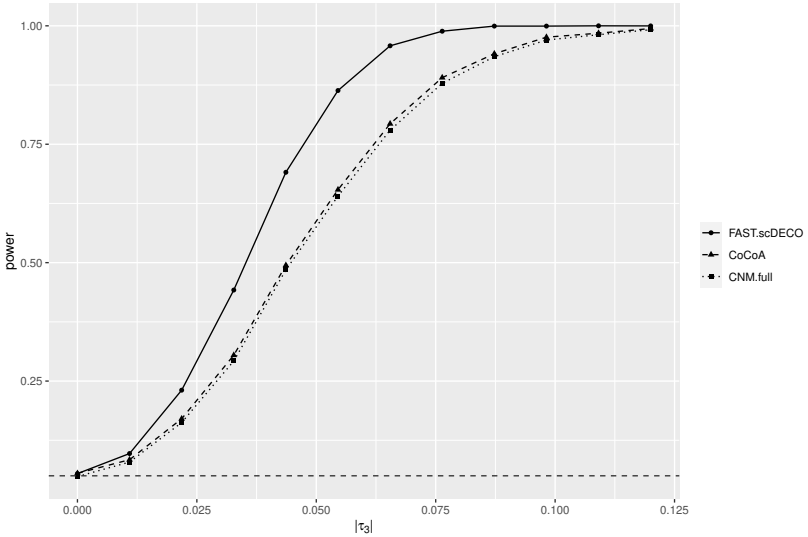


Simulation

	n=1000				n=5000			
	Coverage	MSE	MBE	CI Width	Coverage	MSE	MBE	CI Width
p_1	0.946	0.001	0.021	0.101	0.948	0.000	0.009	0.045
p_2	0.948	0.001	0.018	0.090	0.946	0.000	0.008	0.040
β_{01}	0.945	0.002	0.034	0.168	0.954	0.000	0.015	0.075
β_{02}	0.950	0.001	0.029	0.144	0.952	0.000	0.013	0.064
σ_1	0.947	0.007	0.063	0.305	0.946	0.001	0.028	0.136
σ_2	0.942	0.000	0.007	0.035	0.948	0.000	0.003	0.016
τ_0	0.947	0.000	0.012	0.058	0.953	0.000	0.005	0.026
τ_1	0.948	0.000	0.011	0.056	0.951	0.000	0.005	0.025

- FAST-scDECO takes on average 9.89, 33.39, and 76.52 seconds to fit a samples of size $n = 1000$, $n = 2500$, and $n = 5000$, respectively.
- Compare this to over 2 hours for MCMC methods such as ZENCO.

Simulation


1.00
0.75
0.50
0.25
power
0.000
0.025
0.050
0.075
0.100
0.125
|τ3|
FAST.scDECO
CoCoA
CNM.full

Real Data Analysis

- We analyzed breast cancer scRNA-seq data from GSE176078.
- [Wu SZ et al., 2021] investigated spatially-resolved subclasses of CAFs called iCAFs and myCAFs, while [Papanicolaou et. al., 2022] explored the role of collagen.

Model setup

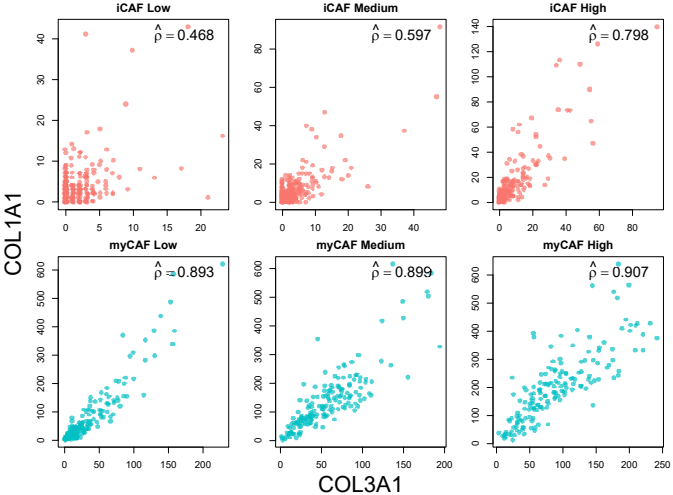
$$
\begin{aligned}
p_1 &= \text{sigmoid}\left\{\kappa_{01} + \kappa_{11}\ \text{iCAF}\right\} \\
p_2 &= \text{sigmoid}\left\{\kappa_{02} + \kappa_{12}\ \text{iCAF}\right\} \\
\mu_1 &= \exp\{\beta_{01} + \beta_{11}\ \text{iCAF} + \beta_{21}\ \text{COL6A3} + \beta_{31}\ \text{iCAF} \times \text{COL6A3}\} \\
\mu_2 &= \exp\{\beta_{02} + \beta_{12}\ \text{iCAF} + \beta_{22}\ \text{COL6A3} + \beta_{32}\ \text{iCAF} \times \text{COL6A3}\} \\
\sigma_1 &= \exp\{\alpha_{01} + \alpha_{11}\ \text{iCAF}\} \\
\sigma_2 &= \exp\{\alpha_{02} + \alpha_{12}\ \text{iCAF}\} \\
\rho &= \tanh\{\tau_0 + \tau_1\ \text{iCAF} + \tau_2\ \text{COL6A3} + \tau_3\ \text{iCAF} \times \text{COL6A3}\}
\end{aligned}
$$

FAST-scDECO's quick computation time makes it possible to conduct a whole-genome search on this dataset.

Real Data Analysis

	Gene 1	Gene 2	τ_1 (myCAF)	τ_1 (iCAF)	$\Delta\tau_1$
1	APOD	SFRP4	0.003 (-0.004,0.011)	0.104 (0.071, 0.136)	0.100 (0.066, 0.134)
2	SFRP4	MT2A	-0.004 (-0.012,0.005)	0.077 (0.038, 0.115)	0.080 (0.041, 0.120)
3	**COL3A1**	**COL1A1**	0.004 (-0.004,0.011)	0.075 (0.056, 0.093)	0.071 (0.051, 0.091)
4	HSPA1B	CFD	-0.002 (-0.014,0.010)	-0.073 (-0.103,-0.043)	-0.071 (-0.103,-0.038)
5	CFD	DNAJB1	0.020 (0.007,0.033)	-0.051 (-0.079,-0.023)	-0.071 (-0.102,-0.040)
6	EEF1A1	IGF2	0.008 (-0.001,0.017)	-0.060 (-0.093,-0.028)	-0.068 (-0.102,-0.034)
7	APOE	RPS9	-0.006 (-0.017,0.006)	0.061 (0.036, 0.085)	0.066 (0.039, 0.094)
8	COL14A1	LUM	0.005 (-0.004,0.015)	0.070 (0.067, 0.073)	0.065 (0.055, 0.075)
9	TMSB10	IGFBP5	-0.001 (-0.012,0.010)	0.062 (0.037, 0.087)	0.063 (0.036, 0.091)
10	CCDC80	LUM	-0.002 (-0.010,0.007)	0.060 (0.037, 0.084)	0.062 (0.037, 0.087)

Real Data Analysis



Conclusion

- We have proposed a powerful, flexible and scalable method for modeling conditional correlation.
- The method accounts for zero-inflation and works with arbitrary marginal distributions– including mixed (continuous and discrete) marginals.
- All parameters can be modeled as functions of covariates.
- Estimation and hypothesis testing are carried out using maximum likelihood.
- Quick computation time makes it feasible to conduct a whole-genome search.

References

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Papanicolaou M, Parker AL, Yam M, Filipe EC et al. (2022)
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